

CONSERVATION LAWS, NEWTON'S LAWS, AND KINEMATICS 2E



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Conservation Laws, Newton's Laws, and Kinematics

Version 2.0

Edited by Christopher Duston

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Kinematics, version 2.0

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CHAPTER OVERVIEW

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This first section is going to serve two purposes: first, introduce some of the key important concepts that we will need right from the beginning, like units and significant figures. Second, we want to open up the question: "what is physics"? Like many such open-ended questions, there are many ways to answer it. It is not going to be our intention to answer this fully in any sense - we just want to answer it enough so that as a student, you know why this book exists and what you should be getting out of it. To get right to the point: **physics is the study of the abstraction of the physical world.**

So that's a pretty obscure statement, so let's break it down a bit. When we say "abstraction", that's basically a way of replacing all the real physical objects out there that we want to understand (balls, cars, airplanes, etc...objects!) with mathematical representations of those objects. These representations are simple - sometimes they are just points, or maybe box-shaped things with wheels to represent cars, etc. The point of the abstraction is both to simplify the problem (points are easier to deal with than real life planes), and also to make it mathematically precise. "how a car moves" is actually a question far too complicated for a simple introduction to the world of physics to handle. There are literally millions (way more!) of interactions taking place in the motion of a car - things like the engine, the wheels, and the transmission - but even little things like the air molecules hitting the outside of the car and slowing it down. By replacing the entire thing with "a point" or "a box with wheels", we make the problem simple enough for us to actually perform some calculations to understand it's motion.

We should be clear what the kind of mathematical precision we are talking about here as well. We don't mean "calculating the exact value of the speed of the car to 5 decimal places" (we'll see in this chapter that kind of precision is not actually what we are after). We mean something more like "numerically well-defined". For example, if we said "The position of the car is 5 m from the end of the race", we can't actually perform any calculations with that information, because *which part of the car are we talking about?* For sure, you could say "the front", or "the back", or "30 cm from the driver", but now we might need to know even more things about the car - how big is it, or how big the seats are (to determine the position of the driver relative to the finish line). The abstraction avoids all this by the replacement of real objects with simple representations of them. If the car is represented as a point, then "5 m from the end of the race" becomes very well-defined, because it can only refer to one location! Similarly for the representation of the car as a box with wheels - more information is required (dimensions of the box and the wheels?), but it's still far less than what might be required for the real, physical car.

Now that we've covered "abstraction", what does "study of the abstraction" mean? Well, it's often said that physics is "the study of the fundamental interactions" - that's what we mean, but we mean it specifically *within the abstraction*. We aren't actually studying how the fundamental interactions of the real world influence real physical objects - we are representing the world with an abstraction, and studying the abstraction. The force of gravity is actually enormously complicated in the real world (it's everywhere, it acts between each individual molecule and each other individual molecule....you'd be calculating for your entire life!), but in our abstraction, it's just a simple force acting between two points. By proposing simple, fundamental interactions between objects we are hoping to model the real world.

Of course, we claim that physics actually is the study of the real world, but you aren't ever going to see that connection by using solely the abstraction - you have to go into the laboratory and perform experiments to verify that our abstraction is an "accurate and precise" representation of the real world. We aren't going to talk much about that in this text, since it is primarily designed to give you an introduction to the prediction and quantitative aspects of the field - hopefully, your theoretical experience with this text will be coupled with a laboratory experience as well, so that you can see how well physics actually does at describing phenomena and events in the real world.

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1.1: About this Text

This text has been written with several goals in mind:

- Break sections into short chunks appropriate for reading before each class meeting, specifically for use in "studio physics" classrooms.
- Present conservation laws before kinematics.
- Acknowledge that students may be co-enrolled in their first calculus course.

Before proceeding with the text, we will motivate and explain each of these goals. Being an open source resource, this text is constantly in a state of improvement. Feel free to write to the editor, Professor Christopher Duston (dustonc at merrimack dot edu) if you find significant errors that you think should be corrected.

This text has been written to support a so-called "studio" approach to teaching physics. There are many flavors of this pedagogy (for example, [SCALE-UP](#) from North Carolina State, [Studio Physics](#) from Florida State, and [Studio Physics](#) at The Massachusetts Institute of Technology), but an essential element is that class time is significantly devoted to problem solving over lecture time. To support this, it is necessary for students to be exposed to the material at some level before even entering the classroom. There are several ways to achieve that (this is the so-called "[flipped classroom model](#)"), but it is our feeling that student success is maximized by designing this initial exposure around the studio model, rather than using a traditional textbook. A traditional and carefully constructed 1000-page physics textbook is a beautiful thing, but if the material is broken into sections appropriate for the content, rather than the delivery system, there will be a fundamental mismatch between the goals of the students and the goals of the textbook. So the first goal of this text is to break the content into sections appropriate for single class meetings. At Merrimack College (where this textbook was designed to be used), there are two class meetings per week, each class meeting is about two hours long, and each meeting tackles one section of this text.

This text was also designed to be the initial source of exposure to the content. Another challenge when using a traditional physics textbook in the studio model is that the first thing a novice reads about might be a completely well-thought out and well-motivated description of a physical law, but a student is likely not prepared for that level of description. The process of learning is not linear and hierarchical, but chaotic and varied. Of course, in the end we want students to be experts *a la* [Bloom's Taxonomy](#), but starting from the bottom and explaining how they get to the top will not service their own internalized development of the material. To that end, the goal of this text is primarily to provide students with enough material to "soak the sponge", so that when they enter the class they are prepared to learn (and make mistakes!), rather than prepared to solve problems on an exam. To achieve this, the material is presented in a streamlined matter, with simplified motivations and only the first few initial steps spelled out. For sure, this means more complicated material is occasionally only briefly touched upon, and in some cases left out completely. However, we would argue that as a fundamental science, more complicated material in the field of physics can always be added in by instructor demand, in a way that their own students can handle it. And, given that anything can be found on the internet, collecting special topics in textbooks seems like something which has already been done by any number of a myriad of authors. If that's what you want, this textbook is not for you!

The content order of this textbook (conservation laws before Newton's laws) is inspired by the wonderful texts of Eric Mazur and Thomas Moore (in fact, the current version of this book blatantly steals the chapter order from Moore's *Six Ideas In Physics*, Units C and N). For more well-thought out motivations, one can consult those admirable texts - for our part, we will simply motivate this in two ways. First, in many ways conservation laws are more fundamental than Newton's laws. Indeed, the argument about the invariance of specific quantities under time and space translations is so fundamental that it borders on the Philosophical nature of the Universe, rather than our particular approach to modeling it. More directly, equations like $\Delta E = 0$ are scalar and discrete, and therefore typically easier for students to deal with than vector expressions that imply continuous change, like $\Sigma \vec{F} = m\vec{a}$. Second, it is our experience that students have already been exposed to Newton's laws and kinematics - and in some cases have built up significant misunderstandings that are difficult to break down. (A trivial example might be the statement that "the force of gravity is negative!", a viewpoint strongly held by many students. This concept is so rife with intellectual inconsistency that I regret even bringing it up...) By starting students with new, consistently correct ideas about modeling and problem solving in the context of conservation laws, we prepare them to tear down their previous knowledge and rebuild it on a more solid foundation.

Finally, this textbook is written with the knowledge that some (many?) students are going to be stepping into their first physics class while simultaneously taking their first calculus class. This breaks the traditional "calculus before physics" mantra that has been a part of our pedagogy for decades. The source of this change will not be covered here, but we are highly motivated to deal with it rather than preach about its evils from the rooftops. Of course, as a textbook that primarily covers mechanics, it's absolutely

true that most of what can be found in this textbook is actually directly derivable from simple ideas based in calculus. However, that is also not usually where students coming to the field are, and it would be an injustice to ask them to climb that mountain before demonstrating the joy of understanding basic concepts in the field. More practically, it is our experience that in so-called "calculus-based physics classes", the biggest challenges for students is not the calculus itself, but the vector algebra and analysis which is utilized in nearly no other field as extensively as physics. Therefore, we do not view the loss of a semester of calculus before this class as a significant barrier, and do not start by assuming students have it. By the second half of the book (Newton's laws), it is assumed that students have a grasp of all the basic notations from calculus.

Open Source Resources and the Construction of This Text

From the outset, we must acknowledge that this text fails in all the goals presented above. Writing a text from scratch is a lengthy business, and we have students to teach! However, since the advent of free and open source educational resources, another avenue has been opened - the "collect, evaluate, and modify" approach. Open source solutions have been available for some time ([University Physics](#) by OpenStax is perhaps the most well-known example, as well as [Calculus-Based Physics](#) by Jeffrey Schnick), but these followed the more traditional order of material. However, once Julio Gea-Banacloche published his fantastic [University Physics I: Classical Mechanics](#), the possibility of satisfying our goals via open source publication became realizable. However, rearranging and breaking apart a textbook comes with it's own dangers, and we are squarely in the middle of those dangers with this text, which was constructed in the following way:

1. Rearrange the Gea-Banacloche text to match the specific order of material, which closely follows Thomas Moore's book.
2. Supplement material which is lacking from OpenStax.
3. Write new material to fill in some gaps.

As such, we owe a great deal of gratitude to both Gea-Banacloche and the OpenStax organization, for making such an approach feasible. However, the responsibility is now all on our shoulders, to continue to modify and develop this book so that eventually we can satisfy the lofty pedagogical and practical goals we outlined at the beginning of this section.

Whiteboard Problems

In our studio classroom, over half the time is spent solving problems on whiteboards in groups. During this time, the instructor (and typically one or two learning assistants) are on hand and available to assist if a group gets stuck. Over the years, we've developed many more of these style problems than we will typically use (3-4 per class period, generally), so we've included the extras in this text. At the end of each chapter we will have an "Example" section, and these whiteboard problems will be presented there (along with some good examples from the two main source texts we discussed above). These can be used identically to example problems, or they could be used as actual whiteboard problems for a studio classroom. In the future, we are planning on adding our entire set of whiteboard problems to these sections - this will correspond with a generally shortening of the actual texts of the chapter, to match our goal of being accessible as a pre-class reading assignment.

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1.2: Modeling in Physics

Classical mechanics is the branch of physics that deals with the study of the motion of anything (roughly speaking) larger than an atom or a molecule. That is a lot of territory, and the methods and concepts of classical mechanics are at the foundation of any branch of science or engineering that is concerned with the motion of anything from a star to an amoeba—fluids, rocks, animals, planets, and any and all kinds of machines. Moreover, even though the accurate description of processes at the atomic level requires the (formally very different) methods of quantum mechanics, at least three of the basic concepts of classical mechanics that we are going to study this semester, namely, momentum, energy, and angular momentum, carry over into quantum mechanics as well, with the last two playing, in fact, an essential role.

Particles in Classical Mechanics

In the study of motion, the most basic starting point is the concept of the *position* of an object. Clearly, if we want to describe accurately the position of a macroscopic object such as a car, we may need a lot of information, including the precise shape of the car, whether it is turned this way or that way, and so on; however, if all we want to know is how far the car is from Fort Smith or Fayetteville, we do not need any of that: we can just treat the car as a dot, or mathematical point, on the map—which is the way your GPS screen will show it, anyway. When we do this, we say that we are describing the car (or whatever the macroscopic object may be) as a **particle**.

In classical mechanics, an “ideal” particle is an object with no appreciable size—a mathematical point. In one dimension (that is to say, along a straight line), its position can be specified just by giving a single number, the distance from some reference point, as we shall see in a moment (in three dimensions, of course, three numbers are required). In terms of energy (which is perhaps the most important concept in all of physics, and which we will introduce properly in due course), an ideal particle has only one kind of energy, what we will later call *translational kinetic energy*; it cannot have, for instance, rotational kinetic energy (since it has “no shape” for practical purposes), or any form of internal energy (elastic, thermal, etc.), since we assume it is too small to have any internal structure in the first place.

The reason this is a useful concept is not just that we can often treat extended objects as particles in an approximate way (like the car in the example above), but also, and most importantly, that if we want to be more precise in our calculations, *we can always treat an extended object (mathematically) as a collection of “particles.”* The physical properties of the object, such as its energy, momentum, rotational inertia, and so forth, can then be obtained by adding up the corresponding quantities for all the particles making up the object. Not only that, but the interactions between two extended objects can also be calculated by adding up the interactions between all the particles making up the two objects. This is how, once we know the form of the gravitational force between two particles (which is fairly simple, as we will see in [Section 8.7](#)), we can use that to calculate the force of gravity between a planet and its satellites, which can be fairly complicated in detail, depending, for instance, on the relative orientation of the planet and the satellite.

The mathematical tool we use to calculate these “sums” is *calculus*—specifically, integration—and you will see many examples of this... in your calculus courses. Calculus I is only a corequisite for this course, so we will not make a lot of use of it here, and in any case you would need multidimensional integrals, which are an even more advanced subject, to do these kinds of calculations. But it may be good for you to keep these ideas on the back of your mind. Calculus was, in fact, invented by Sir Isaac Newton precisely for this purpose, and the developments of physics and mathematics have been closely linked together ever since.

Anyway, back to particles, the plan for this semester is as follows: we will start our description of motion by treating every object (even fairly large ones, such as cars) as a “particle,” because we will only be concerned at first with its translational motion and the corresponding energy. Then we will progressively make things more complex: by considering systems of two or more particles, we will start to deal with the *internal energy* of a system. Then we will move to the study of *rigid bodies*, which are another important idealization: extended objects whose parts all move together as the object undergoes a translation or a rotation. This will allow us to introduce the concept of rotational kinetic energy. Eventually we will consider *wave motion*, where different parts of an extended object (or “medium”) move relative to each other. So, you see, there is a logical progression here, with most parts of the course building on top of the previous ones, and energy as one of the main connecting themes. The technical word for the process being described in the preceding paragraphs is **abstraction**; the essence of which is to take the physical world and model it with abstract mathematical quantities. We can then use these quantities to construct physical theories, make predictions, and verify them with experiments. In this text we will primarily be interested in the middle section of this process - *taking physical laws and making predictions about how objects in the real world will behave*.

Aside: The Atomic Perspective

As an aside, it should perhaps be mentioned that the building up of classical mechanics around this concept of ideal particles had nothing to do, initially, with any belief in “atoms,” or an atomic theory of matter. Indeed, for most 18th and 19th century physicists, matter was supposed to be a continuous medium, and its (mental) division into particles was just a mathematical convenience.

The atomic hypothesis became increasingly more plausible as the 19th century wore on, and by the 1920’s, when quantum mechanics came along, physicists had to face a surprising development: matter, it turned out, was indeed made up of “elementary particles,” but these particles could not, in fact, be themselves described by the laws of classical mechanics. One could not, for instance, attribute to them simultaneously well-defined positions and velocities. Yet, in spite of this, most of the conclusions of classical mechanics remain valid for macroscopic objects, because, most of the time, it is OK to (formally) “break up” extended objects into chunks that are small enough to be treated as particles, but large enough that one does not need quantum mechanics to describe their behavior.

Quantum properties were first found to manifest themselves at the macroscopic level when dealing with thermal energy, because at one point it really became necessary to figure out where and how the energy was stored at the truly microscopic (atomic) level. Thus, after centuries of successes, classical mechanics met its first failure with the so-called *problem of the specific heats*, and a completely new physical theory—quantum mechanics—had to be developed in order to deal with the newly-discovered atomic world. But all this, as they say, is another story, and for our very brief dealings with thermal physics—the last chapter in this book—we will just take specific heats as given, that is to say, something you measure (or look up in a table), rather than something you try to calculate from theory.

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1.3: Units and Standards

Learning Objectives

- Describe how SI base units are defined.
- Describe how derived units are created from base units.
- Express quantities given in SI units using metric prefixes.

As we saw previously, the range of objects and phenomena studied in physics is immense. From the incredibly short lifetime of a nucleus to the age of Earth, from the tiny sizes of subnuclear particles to the vast distance to the edges of the known universe, from the force exerted by a jumping flea to the force between Earth and the Sun, there are enough factors of 10 to challenge the imagination of even the most experienced scientist. Giving numerical values for physical quantities and equations for physical principles allows us to understand nature much more deeply than qualitative descriptions alone. To comprehend these vast ranges, we must also have accepted units in which to express them. We shall find that even in the potentially mundane discussion of meters, kilograms, and seconds, a profound simplicity of nature appears: all physical quantities can be expressed as combinations of only seven base physical quantities.

We define a **physical quantity** either by specifying how it is measured or by stating how it is calculated from other measurements. For example, we might define distance and time by specifying methods for measuring them, such as using a meter stick and a stopwatch. Then, we could define average speed by stating that it is calculated as the total distance traveled divided by time of travel.

Measurements of physical quantities are expressed in terms of **units**, which are standardized values. For example, the length of a race, which is a physical quantity, can be expressed in units of meters (for sprinters) or kilometers (for distance runners). Without standardized units, it would be extremely difficult for scientists to express and compare measured values in a meaningful way (Figure 1.3.1).

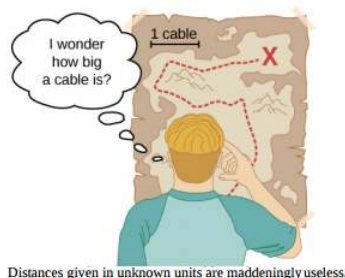


Figure 1.3.1: Distances given in unknown units are maddeningly useless.

Two major systems of units are used in the world: **SI units** (for the French **Système International d'Unités**), also known as the **metric system**, and **English units** (also known as the **customary** or **imperial system**). English units were historically used in nations once ruled by the British Empire and are still widely used in the United States. English units may also be referred to as the **foot–pound–second** (fps) system, as opposed to the **centimeter–gram–second** (cgs) system. You may also encounter the term **SAE units**, named after the Society of Automotive Engineers. Products such as fasteners and automotive tools (for example, wrenches) that are measured in inches rather than metric units are referred to as **SAE fasteners** or **SAE wrenches**.

Virtually every other country in the world (except the United States) now uses SI units as the standard. The metric system is also the standard system agreed on by scientists and mathematicians.

SI Units: Base and Derived Units

In any system of units, the units for some physical quantities must be defined through a measurement process. These are called the **base quantities** for that system and their units are the system's **base units**. All other physical quantities can then be expressed as algebraic combinations of the base quantities. Each of these physical quantities is then known as a **derived quantity** and each unit is called a **derived unit**. The choice of base quantities is somewhat arbitrary, as long as they are independent of each other and all other quantities can be derived from them. Typically, the goal is to choose physical quantities that can be measured accurately to a

high precision as the base quantities. The reason for this is simple. Since the derived units can be expressed as algebraic combinations of the base units, they can only be as accurate and precise as the base units from which they are derived.

Based on such considerations, the International Standards Organization recommends using seven base quantities, which form the International System of Quantities (ISQ). These are the base quantities used to define the SI base units. Table 1.3.1 lists these seven ISQ base quantities and the corresponding SI base units.

Table 1.3.1: ISQ Base Quantities and Their SI Units

ISQ Base Quantity	SI Base Unit
Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Electrical Current	ampere (A)
Thermodynamic Temperature	kelvin (K)
Amount of Substance	mole (mol)
Luminous Intensity	candela (cd)

You are probably already familiar with some derived quantities that can be formed from the base quantities in Table 1.3.1. For example, the geometric concept of area is always calculated as the product of two lengths. Thus, area is a derived quantity that can be expressed in terms of SI base units using square meters ($\text{m} \times \text{m} = \text{m}^2$). Similarly, volume is a derived quantity that can be expressed in cubic meters (m^3). Speed is length per time; so in terms of SI base units, we could measure it in meters per second (m/s). Volume mass density (or just density) is mass per volume, which is expressed in terms of SI base units such as kilograms per cubic meter (kg/m^3). Angles can also be thought of as derived quantities because they can be defined as the ratio of the arc length subtended by two radii of a circle to the radius of the circle. This is how the radian is defined. Depending on your background and interests, you may be able to come up with other derived quantities, such as the mass flow rate (kg/s) or volume flow rate (m^3/s) of a fluid, electric charge ($\text{A} \cdot \text{s}$), mass flux density [$\text{kg}/(\text{m}^2 \cdot \text{s})$], and so on. We will see many more examples throughout this text. For now, the point is that every physical quantity can be derived from the seven base quantities in Table 1.3.1, and the units of every physical quantity can be derived from the seven SI base units.

For the most part, we use SI units in this text. Non-SI units are used in a few applications in which they are in very common use, such as the measurement of temperature in degrees Celsius ($^{\circ}\text{C}$), the measurement of fluid volume in liters (L), and the measurement of energies of elementary particles in electron-volts (eV). Whenever non-SI units are discussed, they are tied to SI units through conversions. For example, 1 L is 10^{-3} m^3 .

Check out a comprehensive source of information on SI units at the National Institute of Standards and Technology (NIST) [Reference on Constants, Units, and Uncertainty](#).

Units of Time, Length, and Mass: The Second, Meter, and Kilogram

The initial chapters in this textmap are concerned with mechanics, fluids, and waves. In these subjects all pertinent physical quantities can be expressed in terms of the base units of length, mass, and time. Therefore, we now turn to a discussion of these three base units, leaving discussion of the others until they are needed later.

The Second

The SI unit for time, the **second** (abbreviated s), has a long history. For many years it was defined as $1/86,400$ of a mean solar day. More recently, a new standard was adopted to gain greater accuracy and to define the second in terms of a nonvarying or constant physical phenomenon (because the solar day is getting longer as a result of the very gradual slowing of Earth's rotation). Cesium atoms can be made to vibrate in a very steady way, and these vibrations can be readily observed and counted. In 1967, the second was redefined as the time required for 9,192,631,770 of these vibrations to occur (Figure 1.3.2). Note that this may seem like more precision than you would ever need, but it isn't—GPSs rely on the precision of atomic clocks to be able to give you turn-by-turn directions on the surface of Earth, far from the satellites broadcasting their location.



Figure 1.3.2: An atomic clock such as this one uses the vibrations of cesium atoms to keep time to a precision of better than a microsecond per year. The fundamental unit of time, the second, is based on such clocks. This image looks down from the top of an atomic fountain nearly 30 feet tall. (credit: Steve Jurvetson)

The Meter

The SI unit for length is the **meter** (abbreviated m); its definition has also changed over time to become more precise. The meter was first defined in 1791 as $1/10,000,000$ of the distance from the equator to the North Pole. This measurement was improved in 1889 by redefining the meter to be the distance between two engraved lines on a platinum–iridium bar now kept near Paris. By 1960, it had become possible to define the meter even more accurately in terms of the wavelength of light, so it was again redefined as 1,650,763.73 wavelengths of orange light emitted by krypton atoms. In 1983, the meter was given its current definition (in part for greater accuracy) as the distance light travels in a vacuum in $1/299,792,458$ of a second (Figure 1.3.3). This change came after knowing the speed of light to be exactly 299,792,458 m/s. The length of the meter will change if the speed of light is someday measured with greater accuracy.

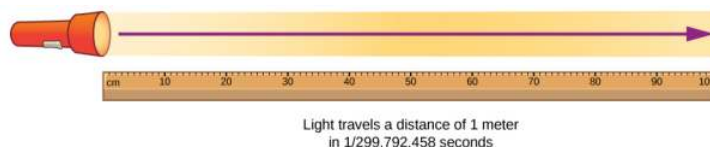


Figure 1.3.3: The meter is defined to be the distance light travels in $1/299,792,458$ of a second in a vacuum. Distance traveled is speed multiplied by time.

The Kilogram

The SI unit for mass is the **kilogram** (abbreviated kg); From 1795–2018 it was defined to be the mass of a platinum–iridium cylinder kept with the old meter standard at the International Bureau of Weights and Measures near Paris. However, this cylinder has lost roughly 50 micrograms since it was created. Because this is the standard, this has shifted how we defined a kilogram. Therefore, a new definition was adopted in May 2019 based on the Planck constant and other constants which will never change in value. We will study Planck’s constant in quantum mechanics, which is an area of physics that describes how the smallest pieces of the universe work. The kilogram is measured on a Kibble balance (see 1.3.4). When a weight is placed on a Kibble balance, an electrical current is produced that is proportional to Planck’s constant. Since Planck’s constant is defined, the exact current measurements in the balance define the kilogram.



Figure 1.3.4: Redefining the SI unit of mass. The U.S. National Institute of Standards and Technology's Kibble balance is a machine that balances the weight of a test mass with the resulting electrical current needed for a force to balance it.

Metric Prefixes

SI units are part of the **metric system**, which is convenient for scientific and engineering calculations because the units are categorized by factors of 10. Table 1.3.1 lists the metric prefixes and symbols used to denote various factors of 10 in SI units. For example, a centimeter is one-hundredth of a meter (in symbols, $1 \text{ cm} = 10^{-2} \text{ m}$) and a kilometer is a thousand meters ($1 \text{ km} = 10^3 \text{ m}$). Similarly, a megagram is a million grams ($1 \text{ Mg} = 10^6 \text{ g}$), a nanosecond is a billionth of a second ($1 \text{ ns} = 10^{-9} \text{ s}$), and a terameter is a trillion meters ($1 \text{ Tm} = 10^{12} \text{ m}$).

Table 1.3.2: Metric Prefixes for Powers of 10 and Their Symbols

Prefix	Symbol	Meaning	Prefix	Symbol	Meaning
yotta-	Y	10^{24}	yocto-	Y	10^{-24}
zetta-	Z	10^{21}	zepto-	Z	10^{-21}
exa-	E	10^{18}	atto-	E	10^{-18}
peta-	P	10^{15}	femto-	P	10^{-15}
tera-	T	10^{12}	pico-	T	10^{-12}
giga-	G	10^9	nano-	G	10^{-9}
mega-	M	10^6	micro-	M	10^{-6}
kilo-	k	10^3	milli-	k	10^{-3}
hecto-	h	10^2	centi-	h	10^{-2}
deka-	da	10^1	deci-	da	10^{-1}

The only rule when using metric prefixes is that you cannot “double them up.” For example, if you have measurements in petameters ($1 \text{ Pm} = 10^{15} \text{ m}$), it is not proper to talk about megagigameters, although $10^6 \times 10^9 = 10^{15}$. In practice, the only time this becomes a bit confusing is when discussing masses. As we have seen, the base SI unit of mass is the kilogram (kg), but metric prefixes need to be applied to the gram (g), because we are not allowed to “double-up” prefixes. Thus, a thousand kilograms (10^3 kg) is written as a megagram (1 Mg) since

$$10^3 \text{ kg} = 10^3 \times 10^3 \text{ g} = 10^6 \text{ g} = 1 \text{ Mg.} \quad (1.3.1)$$

Incidentally, 10^3 kg is also called a **metric ton**, abbreviated t. This is one of the units outside the SI system considered acceptable for use with SI units.

As we see in the next section, metric systems have the advantage that conversions of units involve only powers of 10. There are 100 cm in 1 m, 1000 m in 1 km, and so on. In nonmetric systems, such as the English system of units, the relationships are not as simple—there are 12 in in 1 ft, 5280 ft in 1 mi, and so on.

Another advantage of metric systems is that the same unit can be used over extremely large ranges of values simply by scaling it with an appropriate metric prefix. The prefix is chosen by the order of magnitude of physical quantities commonly found in the task at hand. For example, distances in meters are suitable in construction, whereas distances in kilometers are appropriate for air travel, and nanometers are convenient in optical design. With the metric system there is no need to invent new units for particular applications. Instead, we rescale the units with which we are already familiar.

✓ Example 1.3.1: Using Metric Prefixes

Restate the mass $1.93 \times 10^{13} \text{ kg}$ using a metric prefix such that the resulting numerical value is bigger than one but less than 1000.

Strategy

Since we are not allowed to “double-up” prefixes, we first need to restate the mass in grams by replacing the prefix symbol k with a factor of 10^3 (Table 1.3.2). Then, we should see which two prefixes in Table 1.3.2 are closest to the resulting power of 10 when the number is written in scientific notation. We use whichever of these two prefixes gives us a number between one and 1000.

Solution

Replacing the k in kilogram with a factor of 10^3 , we find that

$$1.93 \times 10^{13} \text{ kg} = 1.93 \times 10^{13} \times 10^3 \text{ g} = 1.93 \times 10^{16} \text{ g}.$$

From Table 1.3.2, we see that 10^{16} is between “peta-” (10^{15}) and “exa-” (10^{18}). If we use the “peta-” prefix, then we find that $1.93 \times 10^{16} \text{ g} = 1.93 \times 10^1 \text{ Pg}$, since $16 = 1 + 15$. Alternatively, if we use the “exa-” prefix we find that $1.93 \times 10^{16} \text{ g} = 1.93 \times 10^{-2} \text{ Eg}$, since $16 = -2 + 18$. Because the problem asks for the numerical value between one and 1000, we use the “peta-” prefix and the answer is 19.3 Pg.

Significance

It is easy to make silly arithmetic errors when switching from one prefix to another, so it is always a good idea to check that our final answer matches the number we started with. An easy way to do this is to put both numbers in scientific notation and count powers of 10, including the ones hidden in prefixes. If we did not make a mistake, the powers of 10 should match up. In this problem, we started with $1.93 \times 10^{13} \text{ kg}$, so we have $13 + 3 = 16$ powers of 10. Our final answer in scientific notation is $1.93 \times 10^1 \text{ Pg}$, so we have $1 + 15 = 16$ powers of 10. So, everything checks out.

If this mass arose from a calculation, we would also want to check to determine whether a mass this large makes any sense in the context of the problem. For this, Figure 1.4 might be helpful.

? Exercises 1.3.1

Restate $4.79 \times 10^5 \text{ kg}$ using a metric prefix such that the resulting number is bigger than one but less than 1000.

Answer

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1.4: Unit Conversion

Learning Objectives

- Use conversion factors to express the value of a given quantity in different units.

It is often necessary to convert from one unit to another. For example, if you are reading a European cookbook, some quantities may be expressed in units of liters and you need to convert them to cups. Or perhaps you are reading walking directions from one location to another and you are interested in how many miles you will be walking. In this case, you may need to convert units of feet or meters to miles.

Let's consider a simple example of how to convert units. Suppose we want to convert 80 m to kilometers. The first thing to do is to list the units you have and the units to which you want to convert. In this case, we have units in meters and we want to convert to kilometers. Next, we need to determine a conversion factor relating meters to kilometers. A **conversion factor** is a ratio that expresses how many of one unit are equal to another unit. For example, there are 12 in. in 1 ft, 1609 m in 1 mi, 100 cm in 1 m, 60 s in 1 min, and so on. Refer to [Appendix B](#) for a more complete list of conversion factors. In this case, we know that there are 1000 m in 1 km. Now we can set up our unit conversion. We write the units we have and then multiply them by the conversion factor so the units cancel out, as shown:

$$80 \cancel{m} \times \frac{1 \text{ km}}{1000 \cancel{m}} = 0.080 \text{ km}. \quad (1.4.1)$$

Note that the unwanted meter unit cancels, leaving only the desired kilometer unit. You can use this method to convert between any type of unit. Now, the conversion of 80 m to kilometers is simply the use of a metric prefix, as we saw in the preceding section, so we can get the same answer just as easily by noting that

$$80 \text{ m} = 8.0 \times 10^1 \text{ m} = 8.0 \times 10^{-2} \text{ km} = 0.080 \text{ km}, \quad (1.4.2)$$

since “kilo-” means 10^3 and $1 = -2 + 3$. However, using conversion factors is handy when converting between units that are not metric or when converting between derived units, as the following examples illustrate.

✓ Example 1.4.1: Converting Nonmetric Units to Metric

The distance from the university to home is 10 mi and it usually takes 20 min to drive this distance. Calculate the average speed in meters per second (m/s). (**Note:** Average speed is distance traveled divided by time of travel.)

Strategy

First we calculate the average speed using the given units, then we can get the average speed into the desired units by picking the correct conversion factors and multiplying by them. The correct conversion factors are those that cancel the unwanted units and leave the desired units in their place. In this case, we want to convert miles to meters, so we need to know the fact that there are 1609 m in 1 mi. We also want to convert minutes to seconds, so we use the conversion of 60 s in 1 min.

Solution

- Calculate average speed. Average speed is distance traveled divided by time of travel. (Take this definition as a given for now. Average speed and other motion concepts are covered in later chapters.) In equation form,

$$\text{Average speed} = \frac{\text{Distance}}{\text{Time}}.$$

- Substitute the given values for distance and time:

$$\text{Average speed} = \frac{10 \text{ mi}}{20 \text{ min}} = 0.50 \frac{\text{mi}}{\text{min}}.$$

- Convert miles per minute to meters per second by multiplying by the conversion factor that cancels miles and leave meters, and also by the conversion factor that cancels minutes and leave seconds:

$$0.50 \frac{\cancel{\text{mile}}}{\cancel{\text{min}}} \times \frac{1609 \text{ m}}{1 \cancel{\text{mile}}} \times \frac{1 \cancel{\text{min}}}{60 \text{ s}} = \frac{(0.50)(1609)}{60} \text{ m/s} = 13 \text{ m/s}.$$

Significance

Check the answer in the following ways:

1. Be sure the units in the unit conversion cancel correctly. If the unit conversion factor was written upside down, the units do not cancel correctly in the equation. We see the “miles” in the numerator in 0.50 mi/min cancels the “mile” in the denominator in the first conversion factor. Also, the “min” in the denominator in 0.50 mi/min cancels the “min” in the numerator in the second conversion factor.
2. Check that the units of the final answer are the desired units. The problem asked us to solve for average speed in units of meters per second and, after the cancelations, the only units left are a meter (m) in the numerator and a second (s) in the denominator, so we have indeed obtained these units.

? Exercise 1.4.1

Light travels about 9 Pm in a year. Given that a year is about 3×10^7 s, what is the speed of light in meters per second?

Answer

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✓ Example 1.4.2: Converting between Metric Units

The density of iron is 7.86 g/cm^3 under standard conditions. Convert this to kg/m^3 .

Strategy

We need to convert grams to kilograms and cubic centimeters to cubic meters. The conversion factors we need are $1 \text{ kg} = 10^3 \text{ g}$ and $1 \text{ cm} = 10^{-2} \text{ m}$. However, we are dealing with cubic centimeters ($\text{cm}^3 = \text{cm} \times \text{cm} \times \text{cm}$), so we have to use the second conversion factor three times (that is, we need to cube it). The idea is still to multiply by the conversion factors in such a way that they cancel the units we want to get rid of and introduce the units we want to keep.

Solution

$$7.86 \frac{\cancel{\text{g}}}{\cancel{\text{cm}^3}} \times \frac{\text{kg}}{10^3 \cancel{\text{g}}} \times \left(\frac{\cancel{\text{cm}}}{10^{-2} \text{ m}} \right)^3 = \frac{7.86}{(10^3)(10^{-6})} \text{ kg/m}^3 = 7.86 \times 10^3 \text{ kg/m}^3$$

Significance

Remember, it's always important to check the answer.

1. Be sure to cancel the units in the unit conversion correctly. We see that the gram (“g”) in the numerator in 7.86 g/cm^3 cancels the “g” in the denominator in the first conversion factor. Also, the three factors of “cm” in the denominator in 7.86 g/cm^3 cancel with the three factors of “cm” in the numerator that we get by cubing the second conversion factor.
2. Check that the units of the final answer are the desired units. The problem asked for us to convert to kilograms per cubic meter. After the cancelations just described, we see the only units we have left are “kg” in the numerator and three factors of “m” in the denominator (that is, one factor of “m” cubed, or “m³”). Therefore, the units on the final answer are correct.

? Exercise 1.4.2

We know from Figure 1.4 that the diameter of Earth is on the order of 10^7 m , so the order of magnitude of its surface area is 10^{14} m^2 . What is that in square kilometers (that is, km^2)? (Try doing this both by converting 10^7 m to km and then squaring it and then by converting 10^{14} m^2 directly to square kilometers. You should get the same answer both ways.)

Answer

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Unit conversions may not seem very interesting, but not doing them can be costly. One famous example of this situation was seen with the **Mars Climate Orbiter**. This probe was launched by NASA on December 11, 1998. On September 23, 1999, while

attempting to guide the probe into its planned orbit around Mars, NASA lost contact with it. Subsequent investigations showed a piece of software called SM_FORCES (or “small forces”) was recording thruster performance data in the English units of pound-seconds ($\text{lb} \cdot \text{s}$). However, other pieces of software that used these values for course corrections expected them to be recorded in the SI units of newton-seconds ($\text{N} \cdot \text{s}$), as dictated in the software interface protocols. This error caused the probe to follow a very different trajectory from what NASA thought it was following, which most likely caused the probe either to burn up in the Martian atmosphere or to shoot out into space. This failure to pay attention to unit conversions cost hundreds of millions of dollars, not to mention all the time invested by the scientists and engineers who worked on the project.

? Exercise 1.4.3

Given that 1 lb (pound) is 4.45 N, were the numbers being output by SM_FORCES too big or too small?

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1.5: Significant Figures

Learning Objectives

- Determine the correct number of significant figures for the result of a computation.
- Describe the relationship between the concepts of accuracy, precision, uncertainty, and discrepancy.
- Calculate the percent uncertainty of a measurement, given its value and its uncertainty.
- Determine the uncertainty of the result of a computation involving quantities with given uncertainties.

Figure 1.5.1 shows two instruments used to measure the mass of an object. The digital scale has mostly replaced the double-pan balance in physics labs because it gives more accurate and precise measurements. But what exactly do we mean by **accurate** and **precise**? Aren't they the same thing? In this section we examine in detail the process of making and reporting a measurement.



Figure 1.5.1: (a) A double-pan mechanical balance is used to compare different masses. Usually an object with unknown mass is placed in one pan and objects of known mass are placed in the other pan. When the bar that connects the two pans is horizontal, then the masses in both pans are equal. The “known masses” are typically metal cylinders of standard mass such as 1 g, 10 g, and 100 g. (b) Many mechanical balances, such as double-pan balances, have been replaced by digital scales, which can typically measure the mass of an object more precisely. A mechanical balance may read only the mass of an object to the nearest tenth of a gram, but many digital scales can measure the mass of an object up to the nearest thousandth of a gram. (credit a: modification of work by Serge Melki; credit b: modification of work by Karel Jakubec)

Accuracy and Precision of a Measurement

Science is based on observation and experiment—that is, on measurements. **Accuracy** is how close a measurement is to the accepted reference value for that measurement. For example, let's say we want to measure the length of standard printer paper. The packaging in which we purchased the paper states that it is 11.0 in. long. We then measure the length of the paper three times and obtain the following measurements: 11.1 in., 11.2 in., and 10.9 in. These measurements are quite accurate because they are very close to the reference value of 11.0 in. In contrast, if we had obtained a measurement of 12 in., our measurement would not be very accurate. Notice that the concept of accuracy requires that an accepted reference value be given.

The **precision** of measurements refers to how close the agreement is between repeated independent measurements (which are repeated under the same conditions). Consider the example of the paper measurements. The precision of the measurements refers to the spread of the measured values. One way to analyze the precision of the measurements is to determine the range, or difference, between the lowest and the highest measured values. In this case, the lowest value was 10.9 in. and the highest value was 11.2 in. Thus, the measured values deviated from each other by, at most, 0.3 in. These measurements were relatively precise because they did not vary too much in value. However, if the measured values had been 10.9 in., 11.1 in., and 11.9 in., then the measurements would not be very precise because there would be significant variation from one measurement to another. Notice that the concept of precision depends only on the actual measurements acquired and does not depend on an accepted reference value.

The measurements in the paper example are both accurate and precise, but in some cases, measurements are accurate but not precise, or they are precise but not accurate. Let's consider an example of a GPS attempting to locate the position of a restaurant in a city. Think of the restaurant location as existing at the center of a bull's-eye target and think of each GPS attempt to locate the

restaurant as a black dot. In Figure 1.5.1a, we see the GPS measurements are spread out far apart from each other, but they are all relatively close to the actual location of the restaurant at the center of the target. This indicates a low-precision, high-accuracy measuring system. However, in Figure 1.5.1b, the GPS measurements are concentrated quite closely to one another, but they are far away from the target location. This indicates a high-precision, low-accuracy measuring system.

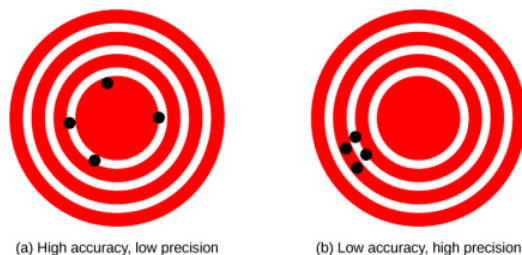


Figure 1.5.2: A GPS attempts to locate a restaurant at the center of the bull's-eye. The black dots represent each attempt to pinpoint the location of the restaurant. (a) The dots are spread out quite far apart from one another, indicating low precision, but they are each rather close to the actual location of the restaurant, indicating high accuracy. (b) The dots are concentrated rather closely to one another, indicating high precision, but they are rather far away from the actual location of the restaurant, indicating low accuracy. (credit a and credit b: modification of works by Dark Evil)

Accuracy, Precision, Uncertainty, and Discrepancy

The precision of a measuring system is related to the **uncertainty** in the measurements whereas the accuracy is related to the **discrepancy** from the accepted reference value. Uncertainty is a quantitative measure of how much your measured values deviate from one another. There are many different methods of calculating uncertainty, each of which is appropriate to different situations. Some examples include taking the range (that is, the biggest less the smallest) or finding the standard deviation of the measurements. Discrepancy (or “measurement error”) is the difference between the measured value and a given standard or expected value. If the measurements are not very precise, then the uncertainty of the values is high. If the measurements are not very accurate, then the discrepancy of the values is high.

Recall our example of measuring paper length; we obtained measurements of 11.1 in., 11.2 in., and 10.9 in., and the accepted value was 11.0 in. We might average the three measurements to say our best guess is 11.1 in.; in this case, our discrepancy is $11.1 - 11.0 = 0.1$ in., which provides a quantitative measure of accuracy. We might calculate the uncertainty in our best guess by using the range of our measured values: 0.3 in. Then we would say the length of the paper is 11.1 in. plus or minus 0.3 in. The uncertainty in a measurement, A , is often denoted as δA (read “delta A ”), so the measurement result would be recorded as $A \pm \delta A$. Returning to our paper example, the measured length of the paper could be expressed as 11.1 ± 0.3 in. Since the discrepancy of 0.1 in. is less than the uncertainty of 0.3 in., we might say the measured value agrees with the accepted reference value to within experimental uncertainty.

Some factors that contribute to uncertainty in a measurement include the following:

- Limitations of the measuring device
- The skill of the person taking the measurement
- Irregularities in the object being measured
- Any other factors that affect the outcome (highly dependent on the situation)

In our example, such factors contributing to the uncertainty could be the smallest division on the ruler is $1/16$ in., the person using the ruler has bad eyesight, the ruler is worn down on one end, or one side of the paper is slightly longer than the other. At any rate, the uncertainty in a measurement must be calculated to quantify its precision. If a reference value is known, it makes sense to calculate the discrepancy as well to quantify its accuracy.

Percent uncertainty

Another method of expressing uncertainty is as a percent of the measured value. If a measurement A is expressed with uncertainty δA , the percent uncertainty is defined as

$$\text{Percent uncertainty} = \frac{\delta A}{A} \times 100\% \quad (1.5.1)$$

✓ Example 1.5.1: Calculating Percent Uncertainty: A Bag of Apples

A grocery store sells 5-lb bags of apples. Let's say we purchase four bags during the course of a month and weigh the bags each time. We obtain the following measurements:

- Week 1 weight: 4.8 lb
- Week 2 weight: 5.3 lb
- Week 3 weight: 4.9 lb
- Week 4 weight: 5.4 lb

We then determine the average weight of the 5-lb bag of apples is 5.1 ± 0.3 lb. What is the percent uncertainty of the bag's weight?

Strategy

First, observe that the average value of the bag's weight, A , is 5.1 lb. The uncertainty in this value, δA , is 0.3 lb. We can use the following equation to determine the percent uncertainty of the weight:

$$\text{Percent uncertainty} = \frac{\delta A}{A} \times 100\% \quad (1.5.2)$$

Solution

Substitute the values into the equation:

$$\text{Percent uncertainty} = \frac{\delta A}{A} \times 100\% = \frac{0.3 \text{ lb}}{5.1 \text{ lb}} \times 100\% = 5.9\% \approx 6\% \quad (1.5.3)$$

Significance

We can conclude the average weight of a bag of apples from this store is $5.1 \text{ lb} \pm 6\%$. Notice the percent uncertainty is dimensionless because the units of weight in $\delta A = 0.3 \text{ lb}$ canceled those in $A = 5.1 \text{ lb}$ when we took the ratio.

? Exercises 1.5.1

A high school track coach has just purchased a new stopwatch. The stopwatch manual states the stopwatch has an uncertainty of ± 0.05 s. Runners on the track coach's team regularly clock 100-m sprints of 11.49 s to 15.01 s. At the school's last track meet, the first-place sprinter came in at 12.04 s and the second-place sprinter came in at 12.07 s. Will the coach's new stopwatch be helpful in timing the sprint team? Why or why not?

Uncertainties in Calculations

Uncertainty exists in anything calculated from measured quantities. For example, the area of a floor calculated from measurements of its length and width has an uncertainty because the length and width have uncertainties. How big is the uncertainty in something you calculate by multiplication or division? If the measurements going into the calculation have small uncertainties (a few percent or less), then the **method of adding percents** can be used for multiplication or division. This method states **the percent uncertainty in a quantity calculated by multiplication or division is the sum of the percent uncertainties in the items used to make the calculation**. For example, if a floor has a length of 4.00 m and a width of 3.00 m, with uncertainties of 2% and 1%, respectively, then the area of the floor is 12.0 m^2 and has an uncertainty of 3%. (Expressed as an area, this is 0.36 m^2 [$12.0 \text{ m}^2 \times 0.03$], which we round to 0.4 m^2 since the area of the floor is given to a tenth of a square meter.)

Precision of Measuring Tools and Significant Figures

An important factor in the precision of measurements involves the precision of the measuring tool. In general, a precise measuring tool is one that can measure values in very small increments. For example, a standard ruler can measure length to the nearest millimeter whereas a caliper can measure length to the nearest 0.01 mm. The caliper is a more precise measuring tool because it can measure extremely small differences in length. The more precise the measuring tool, the more precise the measurements.

When we express measured values, we can only list as many digits as we measured initially with our measuring tool. For example, if we use a standard ruler to measure the length of a stick, we may measure it to be 36.7 cm. We can't express this value as 36.71 cm because our measuring tool is not precise enough to measure a hundredth of a centimeter. It should be noted that the last digit in

a measured value has been estimated in some way by the person performing the measurement. For example, the person measuring the length of a stick with a ruler notices the stick length seems to be somewhere in between 36.6 cm and 36.7 cm, and he or she must estimate the value of the last digit. Using the method of **significant figures**, the rule is that **the last digit written down in a measurement is the first digit with some uncertainty**. To determine the number of significant digits in a value, start with the first measured value at the left and count the number of digits through the last digit written on the right. For example, the measured value 36.7 cm has three digits, or three significant figures. Significant figures indicate the precision of the measuring tool used to measure a value.

Zeros

Special consideration is given to zeros when counting significant figures. The zeros in 0.053 are not significant because they are placeholders that locate the decimal point. There are two significant figures in 0.053. The zeros in 10.053 are not placeholders; they are significant. This number has five significant figures. The zeros in 1300 may or may not be significant, depending on the style of writing numbers. They could mean the number is known to the last digit or they could be placeholders. So 1300 could have two, three, or four significant figures. To avoid this ambiguity, we should write 1300 in scientific notation as 1.3×10^3 , 1.30×10^3 , or 1.300×10^3 , depending on whether it has two, three, or four significant figures. **Zeros are significant except when they serve only as placeholders.**

Significant Figures in Calculations

When combining measurements with different degrees of precision, **the number of significant digits in the final answer can be no greater than the number of significant digits in the least-precise measured value**. There are two different rules, one for multiplication and division and the other for addition and subtraction.

1. **For multiplication and division, the result should have the same number of significant figures as the quantity with the least number of significant figures entering into the calculation.** For example, the area of a circle can be calculated from its radius using $A = \pi r^2$. Let's see how many significant figures the area has if the radius has only two—say, $r = 1.2$ m. Using a calculator with an eight-digit output, we would calculate

$$A = \pi r^2 = (3.1415927...) \times (1.2 \text{ m})^2 = 4.5238934 \text{ m}^2. \quad (1.5.4)$$

But because the radius has only two significant figures, it limits the calculated quantity to two significant figures, or

$$A = 4.5 \text{ m}^2. \quad (1.5.5)$$

although π is good to at least eight digits.

2. **For addition and subtraction, the answer can contain no more decimal places than the least-precise measurement.**

Suppose we buy 7.56 kg of potatoes in a grocery store as measured with a scale with precision 0.01 kg, then we drop off 6.052 kg of potatoes at your laboratory as measured by a scale with precision 0.001 kg. Then, we go home and add 13.7 kg of potatoes as measured by a bathroom scale with precision 0.1 kg. How many kilograms of potatoes do we now have and how many significant figures are appropriate in the answer? The mass is found by simple addition and subtraction:

$$\begin{array}{r} 7.56 \text{ kg} \\ -6.052 \text{ kg} \\ +13.7 \text{ kg} \\ \hline 15.208 \text{ kg} = 15.2 \text{ kg} \end{array} \quad (1.5.6)$$

Next, we identify the least-precise measurement: 13.7 kg. This measurement is expressed to the 0.1 decimal place, so our final answer must also be expressed to the 0.1 decimal place. Thus, the answer is rounded to the tenths place, giving us 15.2 kg.

Significant Figures in This Text

In this text, most numbers are assumed to have three significant figures. Furthermore, consistent numbers of significant figures are used in all worked examples. An answer given to three digits is based on input good to at least three digits, for example. If the input has fewer significant figures, the answer will also have fewer significant figures. Care is also taken that the number of significant figures is reasonable for the situation posed. In some topics, particularly in optics, more accurate numbers are needed and we use more than three significant figures. Finally, if a number is exact, such as the two in the formula for the circumference of a circle, $C = 2\pi r$, it does not affect the number of significant figures in a calculation. Likewise, conversion factors such as 100 cm/1 m are considered exact and do not affect the number of significant figures in a calculation.

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CHAPTER OVERVIEW

2: C2) Particles and Interactions

[2.1: Inertia](#)

[2.2: Momentum](#)

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In this chapter we will study particles and interactions. We will make our first step into the abstraction of the real world by **replacing any objects we want to describe with points**. As we talked about in the last chapter, this is to make things not only 1) simpler, but more importantly 2) mathematical well-defined. The way this works with points is dead simple - take a real object, made up of many (essentially an infinite number of) points, and pick one of it's point to represent the entire thing. This is now well-defined because we can now talk about the position $\vec{r}$ of the object as being exactly the position of the point we chose. And it's far simpler, since we are only dealing with a single point instead of innumerably many.

So after we have a bunch of points, what then? Well, we want to understand the **interactions between the points** (remember, that was one of the ways we described what physics is - the study of these interactions). While we might be tempted to call these interactions "forces", in many cases the forces acting on a particular object are relatively complicated. So first, we are actually going to choose the simplest possible interaction between two objects we can consider - a collision. This is simply when one object comes into contact with another, changing the motion of both.

So what can happen in such a collision? Well, the position $\vec{r}$ and the velocity $\vec{v}$ of either object can change - and that's kind of it, since our objects are just points! The points might have masses, for sure, but for now we are going to assume these masses don't change. (Consider a two-car collision - for sure, some mass is moved back and forth if the collision is bad enough, but generally the two cars keep all their points within each other and don't exchange them.) But how does (for example) the velocity change when the objects interact? It turns out that the velocity is actually not the thing that tells us what happens when two objects collide, it's actually the product of the mass and the velocity, the momentum $\vec{p} = m\vec{v}$. It's pretty intuitive that both the mass and the velocity have to play a role here, since a heavy object hitting a light object is going to be different from two light objects hitting each other. This brings us to our first real principle of physics we are going to study:



Principle of Momentum

Objects interact by transferring momentum

Although this statement seems quite trivial, it actually allows us to perform our first calculations. Consider a collision between two points, each of mass 1 kg, and let's say one of the objects is at rest, while the other is moving towards the first at a speed of 10 m/s. That means the first object has a momentum of 10 kg m/s, while the second has zero (since it's not moving). After they collide, let's say the second object moved away at a speed of 2 m/s. That means it gained 2 kg m/s of momentum...and based on our principle of momentum transfer, the first object must have lost that same 2 kg m/s, leaving it with 8 kg m/s. But that also means we know how fast the object is moving after the collision - since it's mass is 1 kg, it's moving at 8 m/s! This simple example gives us all the essential concepts we are going to be studying not just for this chapter, but for the entire first half of the text.

A brief final note - we didn't really consider anything about the directions that the two objects moved in that example, because it was pretty clear they were moving in a straight line. However, some quantities in physics carry direction, like velocity $\vec{v}$ and position $\vec{r}$ - what about momentum? Well, we defined momentum with a particular formula, $m\vec{v}$. Velocity carries direction, but mass does not, so the momentum will be in the same direction as the velocity. This makes perfect sense with our momentum transfer principle above as well - an object with a particular momentum $\vec{p}$ in a particular direction will transfer that same momentum to the other object, in the same direction.

This chapter is dedicated to a conceptual understanding of this principle - not that there are no calculations, but we are going to be a little careless with things like vectors and directions. We first want to be sure we get some of the conceptual ideas surrounding momentum transfer, and in later chapters we will add in the mathematical formalism required to handle the directions - and all kinds of other interesting collisions!

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